Robin Langer

Agentic Engineer • Claude Code Specialist • PhD Mathematics

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Profile

Mathematician turned agentic engineer. I treat multi-agent systems as an experimental discipline — running controlled sweeps over communication topologies, prompts, and model choices, and measuring how the shape of the agent graph drives system behaviour. I build production tooling on top of Claude Code, orchestrate fleets of autonomous agents with shared memory, and ship ML systems that benchmark favourably against frontier-lab baselines. Co-organiser of *Imagineering with Claude Code* (Melbourne).

Selected Projects

claude-chorus Python • github.com/RaggedR/claude-chorus
Instruction architecture for orchestrating ephemeral and autonomous Claude Code instances with shared culture, coding standards, and persistent memory — a reference design for multi-agent engineering teams.

code-audit Claude Code skills • github.com/RaggedR/code-audit
Production code-auditing toolkit: a swarm of 20–30 audit agents crawls the codebase in configurable topologies, reporting findings back to a coordinator. Includes orchestrator-driven sweeps, two-pass test generation, and component-level analysis, packaged as composable Claude Code skills.

arxiv-intelligence-mcp MCP server • github.com/RaggedR/arxiv-intelligence-mcp
Model Context Protocol server exposing ArXiv search, IBM DocLing PDF parsing, and ChromaDB-backed RAG to Claude Desktop and Cursor. Turns a personal paper library into a tool-callable research assistant.

math-embed (embeddings) Python, PyTorch • github.com/RaggedR/embeddings
Domain-specific embedding model for mathematical paper retrieval, fine-tuned via knowledge-graph contrastive learning. **MRR 0.816 vs. OpenAI's 0.461** on a held-out math-retrieval benchmark.

dreaming-agent Python, Docker • github.com/RaggedR/dreaming-agent
Autonomous research agent that wakes, works, reads, and "dreams" — runs Claude Code in a container with a persistent memory store and a circadian work/consolidate cycle.

Other selected work: nonaga (14-weight GA beating a 570K-parameter AlphaZero net) · blokus (GA-evolved heuristics beat MCTS 8-to-1) · octopus-streams (mech interp of RSK correspondence in Pythia-70m) · melb-tech (knowledge graph of Melbourne's tech community).

Experience

Agentic Engineer | Enspyr Pty Ltd, Melbourne 2017 – Present

- Designed and shipped automated AI research agents driving a Claude-Code-first development workflow.
- Built custom RAG pipelines and domain-specific embedding models for technical document retrieval.
- Audited third-party codebases for production readiness; productised the methodology as reusable Claude Code skills.

Education

PhD, Mathematics — Université Paris-Est (École Doctorale MSTIC), 2013. Thesis: *Cylindric Plane Partitions, Alternating Sign Matrices and Semi-Circular Systems* (arXiv:2110.12629). Advisor: Prof. Philippe Biane.

MSc, Mathematics (First Class Honours) — The University of Melbourne, 2008. Thesis: *Macdonald Polynomials and Symmetric Functions* (arXiv:0907.3950). Advisor: Prof. Ole Warnaar.

BSc Honours, Mathematics (First Class) — The University of Melbourne, 2005.

Publications

R. Langer, C. Turing, and L. Vega. *Composition determines behavior: diversity fingerprints and the strict/lax dichotomy in genetic algorithms.* GECCO Companion '26, San Jose, Costa Rica. ACM, 2026.

References

Dr. Matthieu Josuat-Vergès — CNRS, IRIF (Université Paris-Cité) • matthieu.josuat-verges@irif.fr
Dr. Ekaterina Vassilieva — CNRS, LIX (École Polytechnique) • katya@lix.polytechnique.fr

Skills

Agentic AI: Claude Code, MCP servers, multi-agent orchestration, persistent-memory agent design, prompt engineering.
ML: RAG, embedding models, contrastive fine-tuning, evolutionary algorithms, mechanistic interpretability.
Languages: Python, Dart/Flutter, TypeScript, Haskell, LaTeX.